

Literature Status for arXiv:2412.07920: Sharp Spectral Multipliers Beyond Métivier Groups

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Status

This is a `literature_already_answered` packet. In *Spectral multipliers on Métivier groups*, Niedorf asks whether p -specific spectral multiplier estimates with the sharp threshold

$$s > d \left| \frac{1}{p} - \frac{1}{2} \right|$$

hold beyond the class of Métivier groups [1]. His later paper *Spectral multipliers on two-step stratified Lie groups with degenerate group structure* gives an affirmative answer for explicit, genuinely non-Métivier subclasses [2].

Source Question

Let G be a two-step stratified Lie group of topological dimension d , let L be a sub-Laplacian, and let

$$J_\mu : \mathfrak{g}_1 \longrightarrow \mathfrak{g}_1, \quad \langle J_\mu x, x' \rangle = \mu([x, x']).$$

A Métivier group is characterized by the invertibility of J_μ for every $\mu \neq 0$. After proving the sharp estimate in that nondegenerate setting, arXiv:2412.07920 asks whether the same p -specific regularity threshold holds beyond Métivier groups. It points to weighted Plancherel estimates as the main obstruction when J_μ has a nontrivial kernel.

Later Answer

The introduction of arXiv:2501.16262 explicitly cites the Métivier paper, describes the same question for groups where nondegeneracy fails, and then turns to a class in which J_μ may have a nontrivial kernel. Its Theorem 1.1 proves the following. Suppose the first layer has an orthogonal block decomposition satisfying the paper's Assumptions A and B, and write $\bar{d}_1$ for the total rank of the nonzero spectral blocks. If $\bar{d}_1 \geq d_2 - 1$, then, in its stated range $1 \leq p \leq p_{\bar{d}_1, d_2}$, every bounded Borel function F satisfying

$$\|F\|_{L^2_{s, \text{sloc}}} < \infty, \quad s > d \left(\frac{1}{p} - \frac{1}{2} \right),$$

gives an $L^p(G)$ -bounded operator $F(L)$ for $p > 1$. The corresponding Bochner–Riesz estimate also holds. The supporting paper states explicitly that its theorem is completely new in the non-Métivier case.

The assumptions are verified there for concrete degenerate families. One is a central product of N copies of the free two-step group $N_{3,2}$; for $N \geq 2$ the multiplier conclusion has a nonempty interval with $p > 1$. Another is the Heisenberg–Reiter family $\mathbb{H}_{N,d_2}$ under the stated dimensional condition. In both families the displayed kernels of J_μ show directly that the relevant examples are not Métivier. Hence sharp p -specific multiplier estimates do hold beyond the Métivier class.

Scope

The later theorem does not cover every two-step stratified Lie group. It requires two structural assumptions and a dimensional inequality; its own discussion says that the needed Plancherel estimates are unavailable in general. Thus the existential “beyond Métivier” question has a rigorous affirmative answer for substantial degenerate subclasses, while the universal all-two-step problem remains open.

Provenance Check

The identification is explicit rather than merely inferential: arXiv:2501.16262 cites arXiv:2412.07920 as its Métivier predecessor, restates the question caused by failure of nondegeneracy, and labels its main theorem new in the non-Métivier setting. Both version-2 PDFs are included in this packet.

References

- [1] L. Niedorf, *Spectral multipliers on Métivier groups*, arXiv:2412.07920v2 (2025), arXiv:2412.07920.
- [2] L. Niedorf, *Spectral multipliers on two-step stratified Lie groups with degenerate group structure*, arXiv:2501.16262v2 (2025), arXiv:2501.16262.